

# Observation-space volatility evaluation: measuring proxy reliability and locating where it matters

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## Abstract

Realized variance is measured with error, and the size of that error is normally computed from sampling theory rather than measured. We fit  $\text{Var}(\log RV_M) = c + AM^b$  across sampling grids on ES and NQ futures and on SPY at two venues. The exponent  $b$  lies between  $-0.41$  and  $-0.97$  against a sampling-theory reference of  $-1.13$  to  $-1.21$ ; the reference falls outside the 95 percent bootstrap interval in 18 of 18 cells and in all 54 fits across three proxies. Proxy error decays with sampling frequency more slowly than theory requires. We read reliability from the fitted intercept, which assumes nothing about that decay, and use thirteen pre-registered kill conditions to separate the decisions it changes from those it does not. Volatility-target sizing moves by 1.008 percent; the variance-to-volatility convexity adjustment is overstated by 5.9 to 134.3 percent.

## 1 Introduction

Every evaluation of a volatility forecast, every reliability correction to an information coefficient, and every model comparison conducted on a realized loss [Patton, 2011, Hansen et al., 2011] takes place in an observation space whose coordinates carry measurement error, since realized variance only estimates integrated variance. In applied practice, however, the magnitude of that error is almost never measured; it is computed from a sampling-theory formula and thereafter treated as known.

The quantity at issue is the reliability coefficient

$$\lambda = \frac{\text{Var}(\log IV)}{\text{Var}(\log RV_M)}, \quad (1)$$

which measures the share of observed dispersion in the log proxy attributable to the latent integrated variance. Wherever  $\lambda$  appears in the applied literature, the denominator of (1) is not estimated from the data but supplied by theory, the value supplied being the sampling variance predicted for a proxy built from  $M$  returns, which is  $2/M$  asymptotically [Barndorff-Nielsen and Shephard, 2002], or  $\psi'(M/2)$  exactly under Gaussian returns and constant within-window volatility. That substitution predicts how proxy error behaves without measuring how it behaves, and the prediction can be tested against the sampling grid.

The substitution does not survive the test. Fitting

$$\text{Var}(\log RV_M) = c + AM^b \quad (2)$$

across sampling grids yields  $b$  between  $-0.41$  and  $-0.97$ , whereas fitting the same functional form to  $\psi'(M/2)$  over the identical grid points returns a reference between  $-1.13$  and  $-1.21$ . The reference falls outside the 95 percent bootstrap interval on  $b$  in 18 of 18 cells, and in all 54 fits obtained when realized variance is replaced by the flat-top realized kernel of Barndorff-Nielsen et al. [2008] or the two-scale estimator of Zhang et al. [2005]. Proxy error does decay as the sampling frequency rises, but not at the rate sampling theory requires, and the median gap between the fitted exponent and its reference is 9.6 times the median standard error on the exponent.

With the denominator substitution unavailable, reliability has to be read from the data and not imposed upon them. We estimate  $\lambda$  from the fitted intercept of (2), which requires no assumption about the rate at which proxy error decays, and we report the cost of that choice alongside its motivation. The intercept route violates the unit interval in 4 of 128 grid rows, against 14 for the subsample estimator and 0 for the asymptotic-variance estimator, so it is better behaved than one conventional alternative and worse behaved than the other.

Having measured  $\lambda$ , we ask where the measurement changes a decision, and we separate the cases using the thirteen pre-registered kill conditions of Table 2, each with its threshold fixed before the measurement it adjudicates. Proxy noise proves to be second-order wherever the loss surface is smooth or the contamination averages away over the estimation window, as it does for volatility-target sizing, which moves by at most 1.008 percent in relative tracking error, and for inverse-volatility risk parity at monthly rebalancing, which moves by 0.0038 percent. Proxy noise is first-order, by contrast, wherever the quantity of interest depends on the variance of the estimate and additional data does not reduce the contamination, and there the variance-to-volatility convexity adjustment is overstated by between 5.9 and 134.3 percent, while a regime threshold placed at the median misclassifies between 7.9 and 13.6 percent of days.

Beyond the question of magnitude, we establish two settings in which the measured coefficient cannot be inserted at all. Within-window standardisation annihilates any affine correction to an observable exactly, so no such correction survives the transformation that regime classifiers routinely apply, and a known state-independent noise variance entered into the emission of a two-state model adds no discrimination. Both are negative results about insertion, not about measurement.

## 2 Data and sample construction

### 2.1 Instruments and sample

The futures evidence is ES and NQ one-minute OHLCV bars from Databento (GLBX.MDP3), running from January 2016 through December 2023 by CME trade date, with everything dated 2024-01-01 onward held out. The extract carries 11,318,126 rows before filtering and 5,179,550 after; front-month selection is by daily volume, and calendar spreads are removed at the symbol level, which accounts for 1,031,774 rows. After the exclusions set out below, the estimation sample stands at 1,901 sessions per root on regular trading hours, and at 1,953 and 1,948 sessions on the full Globex session for ES and NQ respectively.

Each instrument is evaluated under two session geometries, RTH covering the 09:30–16:00 New York day session with 390 one-minute bars and GLOBEX the full 18:00–17:00 session with 1,380. The two are not nested samples of a single process, since the Globex window opens into Asian hours and carries the overnight leg, and we report both throughout without selecting between them.

The cross-sectional evidence is SPY at one-second resolution from two venues, NYSE Arca

(ARCX.PILLAR) and Nasdaq (XNAS.ITCH), covering May 2018 through December 2023. Of the 1,427 regular-hours sessions in the raw extract, 1,415 survive the filters and enter the analysis. Neither venue is the consolidated tape, since Arca and Nasdaq together carry roughly a third of consolidated SPY volume, and a single-venue realized variance is computed from a subset of the trades determining the price. We never pool the venues but report each separately, so a result holding on both is a result holding on two independent subsets and not on one aggregate.

## 2.2 Sampling grids

Realized variance is computed on sub-bar grids within each window, this being the sampling dimension along which signature plots are conventionally read [Andersen et al., 2000]. The grids are  $M \in \{5, 6, 10, 13, 26, 78, 195, 389\}$  for RTH daily,  $\{5, 6, 10, 12, 23, 46, 138, 276, 345, 1379\}$  for Globex daily,  $\{4, 5, 6, 10, 12, 15, 20, 30, 60\}$  at the hourly horizon and  $\{5, 6, 10, 15, 30\}$  at the thirty-minute horizon, while for SPY the grid runs from  $M = 5$  to  $M = 23,400$  at one-second sampling. The five-minute equivalent, which is the conventional choice and the one we report as headline, is  $M = 78$  on RTH daily,  $M = 276$  on Globex daily,  $M = 12$  hourly and  $M = 6$  at thirty minutes.

The thirty-minute grid carries only five points against the three parameters of (3); fits obtained on it are reported below, and the admissibility screen of Section 3 excludes them on that ground.

## 2.3 Session exclusions

Sessions are removed by four rules, every one of which is a calendar or data-presence rule. Roll sessions and their immediate neighbours are excluded, at 96 sessions per root per geometry, while early closes are excluded on RTH and, on Globex, only where the overnight leg is also incomplete, since a 13:00 halt leaves the Globex window substantially intact. That gives 68 sessions per root on RTH against 16 and 21 on Globex. Exchange holidays are taken from the published CME equity-index calendar and generated by rule, while exchange-declared halts, including the five March 2020 circuit-breaker sessions, are handled by excluding those minutes in which the exchange printed no bar.

No exclusion is conditioned on a realized quantity, because the object being estimated is the dispersion of log realized variance, and dropping sessions on the ground that their realized variance is unusual would remove the very observations that identify that dispersion. An earlier revision of this work applied a blanket halt-to-close rule which removed windows carrying non-zero realized variance, and those windows were among the highest-volatility sessions in the sample. Under the data-presence rule adopted here, the largest number of excluded windows with non-zero realized variance in any cell is 1.

# 3 The proxy-error scaling exponent

## 3.1 Specification and estimation

For a window divided into  $M$  sub-bars, realized variance is  $RV_M = \sum_{i=1}^M r_i^2$ , whose logarithm carries the log integrated variance together with an error that shrinks as  $M$  grows. Across a grid of sampling frequencies, the observed dispersion of  $\log RV_M$  should then decompose into a component

that does not depend on the grid and a component that vanishes along it. We fit that decomposition directly,

$$\text{Var}(\log RV_M) = c + A M^b, \quad (3)$$

by nonlinear least squares over the grid points of each cell, taking the variance at each  $M$  across the sessions of the estimation sample. The intercept  $c$  is the  $M \rightarrow \infty$  limit and estimates  $\text{Var}(\log IV)$ , while the term  $A M^b$  measures what the choice of sampling grid contributes to the dispersion actually observed. Throughout, a cell denotes one instrument observed under one session geometry, one boundary treatment and one horizon, which gives sixteen futures cells alongside the two SPY venues, eighteen fits in all.

### 3.2 The sampling-theory reference

Under Gaussian returns and volatility held constant within the window,  $M \cdot RV_M / IV$  is distributed as  $\chi_M^2$ , so that

$$\text{Var}(\log RV_M) = \psi'(M/2), \quad (4)$$

where  $\psi'$  is the trigamma function, equal to  $2/M$  to leading order [Barndorff-Nielsen and Shephard, 2002]. The exponent implied by (4) is not  $-1$ , because the leading-order approximation is not what a finite grid delivers. Fitting (3) to (4) over the same grid points a cell actually uses returns  $-1.1444$  on the RTH daily grid,  $-1.1386$  on Globex daily,  $-1.2097$  at the hourly horizon and  $-1.1971$  at thirty minutes. We use the grid-matched reference throughout, and every comparison reported below is between two fits of the same functional form taken over the same points.

Where the data satisfy the assumption behind (4), the estimator recovers that reference. A synthetic arm with i.i.d. lognormal integrated variance and Gaussian innovations, passed through the identical aggregation and fitting code, returns  $-1.1850$  on Globex daily,  $-1.2451$  on RTH daily,  $-1.1928$  hourly and  $-1.2168$  at thirty minutes. Each of these sits within seed dispersion of the analytic value for its grid, which establishes that the code path does not itself manufacture the departure reported below.

### 3.3 Fitted exponents

Table 1: Fitted exponent by cell, with the bootstrap interval from 2,000 joint resamples over sessions and the grid-matched reference. Boundary treatments B0 and B1 give identical fits and are shown once. RMSE and the condition number describe the fit at each cell. Read from `sessions/s10-exponent-audit/results/phase1_bootstrap.csv`.

| Cell           | $b$     | 95% interval       | Reference | RMSE   | Condition |
|----------------|---------|--------------------|-----------|--------|-----------|
| ES/GLOBEX 1day | -0.4427 | [-0.5508, -0.3398] | -1.1377   | 0.0499 | 25.0      |
| NQ/GLOBEX 1day | -0.6886 | [-0.8022, -0.5780] | -1.1377   | 0.0365 | 39.0      |
| ES/RTH 1day    | -0.6274 | [-0.7558, -0.4982] | -1.1444   | 0.0291 | 36.5      |
| NQ/RTH 1day    | -0.9744 | [-1.1255, -0.8353] | -1.1444   | 0.0162 | 70.5      |
| ES/RTH 1h      | -0.4658 | [-0.5345, -0.4004] | -1.2097   | 0.0151 | 37.4      |
| NQ/RTH 1h      | -0.8032 | [-0.8944, -0.7150] | -1.2097   | 0.0119 | 48.4      |
| ES/RTH 30min   | -0.4111 | [-0.4843, -0.3422] | -1.1971   | 0.0045 | 97.8      |
| NQ/RTH 30min   | -0.7010 | [-0.7976, -0.6092] | -1.1971   | 0.0053 | 73.9      |
| SPY/ARCX tick  | -0.5333 | [-0.6807, -0.3896] | -1.1262   | 0.0558 | 23.2      |
| SPY/XNAS tick  | -0.5985 | [-0.7628, -0.4468] | -1.1262   | 0.0590 | 27.3      |

Table 1 reports the fitted exponent, its bootstrap interval and the grid-matched reference for every cell, and Figure 1 plots the same fits against  $M$  on log axes. Across the futures cells the exponent lies between  $-0.4111$  and  $-0.9744$ , and the two SPY venues return  $-0.5333$  and  $-0.5985$ . The grid-matched reference falls outside the 95 percent bootstrap interval in 18 of 18 cells. Whether a separation of that size is large relative to the precision of the fit is settled by comparing the two directly, and the median standard error on  $b$  is 0.0536 against a median gap to the reference of 0.5169, a ratio of 9.6. Proxy error does decay as the sampling frequency rises, but it does not decay at the rate sampling theory requires, and the shortfall is roughly an order of magnitude larger than the uncertainty attaching to any individual fit.

### 3.4 Admissibility of the fits

Three parameters fitted to a short grid can return an exponent that is an artifact of the fit and not a feature of the data. We admit a fit only where  $A > 0$ ,  $b < 0$ , the grid carries at least twice as many points as the model has parameters, and  $|b| > 0.01$ . Of these four conditions, the two governing grid length and magnitude are the ones that bind. Of 18 fits on the extended grids, 14 pass, and these comprise 12 futures cells, being six distinct cells under two boundary treatments that give identical fits, together with both SPY venues. The four exclusions are the thirty-minute cells, whose grids carry only five points. Under the sign-only screen used before this one, 26 of 34 fits pass, among them a Globex daily fit on the restricted grid with  $b = -1.23 \times 10^{-4}$  and a condition number of  $5 \times 10^{11}$ , which the conditions on point count and magnitude remove.

Intercept and exponent are strongly correlated in (3), and over the bootstrap draws the correlation between  $c$  and  $b$  runs from  $-0.065$  to  $-0.973$ , with the worst values arising in the five-point grids. The reported interval on  $b$  already carries that coupling, since the bootstrap resamples the sessions and refits all three parameters jointly rather than profiling one at a time.

### 3.5 Candidate explanations for the departure

Several explanations for the departure are available that do not require abandoning sampling theory, and each is tested in turn below.

**Grid dependence:** Dropping each grid point in turn and refitting moves  $b$  by a median of 0.019 and by at most 0.232, against a median gap to the reference of 0.5169, so no single point in the grid carries the result. The most influential point is an endpoint of the grid in all 18 cells, which is the pattern a power law produces, since the extreme frequencies exert the greatest leverage on a fitted slope.

**Pooling across calendar years:** Fitting within each calendar year returns an exponent steeper than the pooled one in every cell, by a mean of 0.176, so part of the departure is attributable to pooling heterogeneous years. Pooling accounts for a mean 36.0 percent of the pooled-to-reference gap, with a range of 23.6 to 61.1 percent across cells. That leaves a residual of 0.357 lying in the same direction in every cell.

**Alternative proxies:** Fitting (3) to the flat-top realized kernel of Barndorff-Nielsen et al. [2008] and to the two-scale estimator of Zhang et al. [2005] on identical windows gives mean exponents of

$-0.5214$  and  $-0.4953$  respectively, against  $-0.6311$  for realized variance. The noise-robust proxies are on average flatter, not steeper; across all 54 fits the reference lies inside the bootstrap interval 0 times, and removing microstructure noise by two independently published methods does not recover the sampling-theory exponent.

**Heavy tails and localized features:** Replacing the Gaussian innovation with a Student- $t$  standardised to unit variance, at the tail index measured on this sample, moves the exponent from  $-1.2091$  to  $-1.1420$  on Globex and from  $-1.1847$  to  $-1.0885$  on RTH, which is within seed dispersion and nowhere near the observed range. A single dominant return placed at a fixed position inside the window leaves the exponent between  $-1.0743$  and  $-1.1752$  wherever that position is set. The whole class of explanation can in any case be retired analytically, since a feature carrying share  $s$  of realized variance contributes an  $M$ -invariant floor of  $2s^2$ , and supplying the measured excess would then require  $s$  between 16.1 and 32.3 percent against shares running from 0.41 to 4.04 percent, short by a factor of 4.0 to 71.5.

One mechanism does move the exponent into the observed range. Within-window volatility dispersion, calibrated to the measured ratio of realized quarticity to squared realized variance, reproduces the observed exponent on Globex geometry at every Hurst index tested, and on RTH geometry at some but not all of them. The mechanism is established for one session geometry and only partial for the other. The roughness of the within-window path, as distinct from its dispersion, moves the exponent by less than its own between-seed dispersion and cannot account for the residual.

We are less confident in the calibration than in the arm. It matches a single moment ratio, and the mapping from that ratio to a dispersion is a choice of specification rather than a measurement, so a different within-window volatility process matching the same ratio need not reproduce the exponent. The dispersion figures are outputs of the calibration and not observations, since the volatility path within a window is never observed. We take the direction as established and the magnitude as conditional on the parameterisation.

## 4 The intercept estimator

The scaling result of Section 3 removes the assumption on which the conventional reliability estimators rest. Where  $\text{Var}(\log RV_M)$  fails to decay at the rate sampling theory requires, an estimator that substitutes the theoretical sampling variance for the measured one substitutes a quantity the data contradict. We read reliability off the fitted intercept directly, and devote the rest of this section to what that choice costs.

Writing the fit of (3) as  $\text{Var}(\log RV_M) = c + AM^b$ , the intercept  $c$  is the  $M \rightarrow \infty$  limit and estimates  $\text{Var}(\log IV)$ . That suggests the definition

$$\lambda_M = \frac{c}{\text{Var}(\log RV_M)}, \quad (5)$$

for the share of observed log-variance dispersion attributable to the integrated variance and not to the proxy. Equation (5) invokes no assumption about how proxy error scales in  $M$ , drawing only on the fitted limit and on the variance observed directly at the grid point of interest. That narrow property is the whole of its claim to preference over the alternatives.

Two conventional estimators are available for comparison, the first forming  $\lambda$  from contiguous half-window subsamples and the second the asymptotic variance of the truncated estimator keyed to realized quarticity, both bounded in  $(0, 1]$  by construction. Across the 128 (cell,  $M$ ) rows of the extended grid the subsample estimator falls outside those bounds 14 times, the asymptotic-variance estimator 0 times, and equation (5) 4 times. On the only criterion available without knowledge of the truth, then, the intercept route is better behaved than one conventional estimator and worse behaved than the other. We record that ranking and do not select on it.

## 4.1 Identifiability

In the functional form of (3) the intercept and the exponent are strongly correlated, and over the bootstrap draws the correlation between  $c$  and  $b$  ranges from  $-0.065$  to  $-0.973$ , with the asymptotic correlation at the optimum running from  $-0.748$  to  $-0.993$ . The worst values arise in the five-point grids, where three parameters are fitted to five observations. The bootstrap interval on  $b$  already carries that coupling, since it resamples the sessions and refits all three parameters jointly. The median standard error on  $b$  is 0.054, against a median gap to the reference of 9.6 times that figure. Coupled parameters widen an interval without displacing it, so the coupling introduces no bias in  $b$ . It does mean that  $c$  is not estimated independently of the shape assumed for the decay, and every  $\lambda$  reported below inherits that dependence.

The tightened admissibility screen of Section 3 requires at least twice as many grid points as parameters, together with  $|b| > 0.01$ ,  $A > 0$  and  $b < 0$ . It admits 14 of 18 fits, against 26 of 34 under the sign-only screen that preceded it, the four exclusions being the thirty-minute cells with their five grid points. On the restricted grid the screen admits nothing at all, and  $\lambda$  is undefined at all four RTH intraday cells, because a three-parameter fit there is left with a single grid point.

## 4.2 Composition of $AM^b$

Equation (5) treats the entire excess of  $\text{Var}(\log RV_M)$  over  $c$  as proxy error, which is an assumption the data do not support cleanly. Proxy error can also be measured directly, as the variance of  $\log RV_M$  minus the log of a finest-grid flat-top realized kernel [Barndorff-Nielsen et al., 2008] computed on the same windows. That measurement gives between 9.1 and 41.8 percent of the fitted  $AM^b$  at the same grid point, smaller by a factor of 2.4 to 11.

Neither figure bounds the truth on its own. The fitted term is an upper bound on the noise, because it absorbs any component of the excess that belongs to the price process itself and is not error in measuring that process, and Section 3 establishes within-window volatility dispersion as one such component. The kernel difference is a lower bound, because the kernel is computed from the same window and the same price path, so its error is positively correlated with the proxy's and the difference between them understates both. The two constructions bracket  $\lambda$ , and the bracket does not close.

The benign reading of that gap is that a common scaling separates the two measurements, and that they would agree once the kernel's own error were accounted for. The cell-by-cell pattern does not support that reading, since the ratio varies by a factor of 4.6 across the four cells, from 0.091 at ES/GLOBEX to 0.418 at NQ/RTH, and is ordered by the very reliability the estimator is attempting to measure. A common scaling would not produce that ordering. We report  $\lambda$  from the fitted term throughout, state plainly that the fitted term is an upper bound on the noise share and the reported  $\lambda$  therefore a lower bound, and leave the composition unresolved.

## 5 Kill conditions

Thirteen conditions were pre-registered, each with its threshold fixed before the measurement it adjudicates and its null abstract drafted before any result was seen. Table 2 gives all thirteen. A condition fires when the null holds, which is to say when the measured quantity falls inside the threshold and reliability does not change the decision.

Two of the thirteen concern the measurement itself. K2 asks whether any reliability estimator is grid-invariant and fires, in the sense that none is: the ratio of largest to smallest implied  $\text{Var}(\log IV)$  across a grid runs from 1.05 for the best estimator to 1.97 for the worst. K3 asks whether the scaling result survives its own uncertainty and stands, with the reference inside the interval in 0 of 54 fits. The remaining eleven are applications.

### 5.1 What separates them

Proxy noise is second-order where the loss surface is smooth in the corrected quantity, or where the contamination averages away as the estimation window lengthens. It is first-order where the quantity of interest depends on the variance of the estimate rather than its level, and where more data therefore does not reduce the contamination. The distinction is not about the size of  $\lambda$ , which is the same in both cases, and it predicts which side a new application falls on.

### 5.2 The second-order cases

Volatility-target sizing is quadratic in the shrinkage weight near its optimum, so a first-order error in  $\lambda$  produces a second-order loss: replacing the textbook weight with the measured one moves out-of-sample tracking error by at most 1.008 percent against a 5 percent threshold, in the same direction in every cell, at every point of an eightfold cost sweep. Inverse-MSE combination weights move by 0.50 percent in the same metric, because the proxy noise variance is common to every model in the set and cancels from the relative weights to first order. Inverse-volatility risk parity moves the allocation by a mean absolute weight deviation of 0.000145 against a 0.02 threshold, and the reason generalises: the bias in  $\mathbb{E}[1/\hat{\sigma}]$  scales as one over the averaging window, so monthly rebalancing on a twenty-one-session estimate divides it by twenty-one. Risk limits fire on the stop-out at a spurious breach rate of 0.966 percent against 1 percent, and we report the leverage cap as untested rather than as passing, since it bound in 0 of 2,524 decision points and would first bind at 1.434 times a ten percent target.

### 5.3 The first-order cases

The variance-to-volatility convexity adjustment depends on  $\text{Var}(\log IV)$  directly, and substituting the proxy overstates it by between 5.9 and 134.3 percent, which exceeds its 5 percent threshold in every cell. Averaging over more windows improves the estimate of a contaminated population quantity and does not decontaminate it. A regime threshold placed at the median has no curvature to absorb the error, and misclassification runs between 7.9 and 13.6 percent against a 5 percent threshold. Proxy noise in the estimation of the Hurst exponent enters as a nugget that does not

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<sup>2</sup>The session record uses overlapping labels. Item 61 relabelled the MCS-composition condition K2 while the specification already used K2 for grid invariance, and two distinct conditions were both called K3. The paper rennumbers K1–K13 in the order shown; the repository retains the original labels, given here in the second column.

Table 2: Kill conditions and their determinations.<sup>2</sup> Generated from the determination artifacts listed in `paper/k_table.csv`; the margin column gives the measured quantity against its pre-registered threshold, and the final column records whether the application was selected before or after the first-order criterion was articulated.

| Paper | Session      | Content                                                   | Determination                | Margin                                                                     | Selected |
|-------|--------------|-----------------------------------------------------------|------------------------------|----------------------------------------------------------------------------|----------|
| K1    | K1/K2        | reliability correction does not change MCS composition    | INDETERMINATE                | excess 20.8pp on clean geometry, not tracking lambda                       | before   |
| K2    | K2 (grid)    | no reliability estimator is grid-invariant                | FIRES                        | lambda*Var(log RV) max/min ratio 1.05 best, 1.97 worst                     | before   |
| K3    | K3 (scaling) | proxy-error scaling inconsistent with sampling theory     | STANDS                       | reference inside the 95% interval in 0 of 54 proxy fits                    | before   |
| K4    | K3 (sizing)  | sizing consequence null                                   | FIRES                        | max R2-vs-R3 relative TE difference 1.008% vs 5%                           | before   |
| K5    | K4           | risk-limit breaches                                       | stop-out FIRES; cap UNTESTED | spurious 0.966% vs 1%; cap bound 0 of 2524                                 | before   |
| K6    | K5           | inverse-MSE combination weights                           | FIRES                        | max relative TE difference 0.50% vs 5%                                     | before   |
| K7    | K6           | variance-to-volatility convexity adjustment               | DOES NOT FIRE                | overstatement 5.9-134.3% vs 5%                                             | before   |
| K8    | K7           | inverse-volatility risk parity                            | FIRES                        | mean  dw  0.000145 vs 0.02                                                 | before   |
| K9    | K8           | regime misclassification at a median threshold            | DOES NOT FIRE                | analytic 7.9-13.6% vs 5%                                                   | after    |
| K10   | K9           | HAR persistence attenuation                               | INDETERMINATE                | daily shift 4.8-116% vs 10%, spanning the threshold on the Sigma_E choice  | after    |
| K11   | K10          | Hurst exponent bias from proxy noise                      | DOES NOT FIRE                | shift 0.026-0.213 vs 0.02 on a fixed lag window                            | after    |
| K12   | K11          | regime classification under an observable-side correction | DOES NOT FIRE                | 0.00pp from the correction; effect is A3 not being A2                      | after    |
| K13   | K12          | measurement error in the emission                         | FIRES                        | 0.00pp maximum reduction at every scaling, floor binding in 63% of windows | after    |

vanish at short lags, and correcting for the measured  $\lambda$  shifts  $H$  by between 0.026 and 0.213 on a fixed lag window against a 0.02 threshold.

#### 5.4 The two indeterminate conditions

K1 asks whether the reliability correction changes model confidence set composition. Composition differs in 20.8 percent of comparisons on the clean geometry after subtracting a placebo scheme that conditions on a variable with no proxy content, so the placebo accounts for roughly seventy percent of the raw rate and what survives it is the remaining twenty-one points. What remains does not track  $\lambda$  across horizons, running 12.5, 43.8 and 6.3 percent as reliability falls from 1.001 to 0.856 to

0.713, which is not the ordering the mechanism predicts. We report it as indeterminate rather than as either result. K10 asks whether the attenuation of HAR persistence coefficients is material. The daily coefficient shifts by between 4.8 and 116 percent depending on which construction supplies the measurement-error covariance, and the two constructions bracket the 10 percent threshold from either side. The bracket is the same one that leaves the composition of  $AM^b$  unresolved in Section 4.2, resurfacing inside an application.

## 5.5 The criterion’s record

Eight conditions were selected before the criterion of Section 5.1 was articulated and five of them fire. Five were selected after it and three return material effects. We note the ordering because the criterion was derived from the first eight and tested on the last five, not because five of thirteen is a rate worth reporting. Two of the five later conditions test whether the measured coefficient can be inserted at all rather than whether it matters, and Section 6 takes those up separately.

## 6 Insertion into a regime classifier

A measured reliability is of use only where it can enter a decision, and we test the two points at which it can enter a regime classifier built on a rolling two-state Gaussian hidden Markov model, in the specification of Blake et al. [2025]. Neither insertion works, and the two failures share a common cause that has nothing to do with the size of  $\lambda$ .

### 6.1 Shrinkage of the observable

The natural insertion is to shrink the observable toward its unconditional mean with weight  $\lambda$ , replacing  $\log RV_M$  with  $(1 - \lambda)\mu + \lambda \log RV_M$  before classification. The specification standardises the observable within each rolling window, and for any window  $W$  the shrunk series has mean  $(1 - \lambda)\mu + \lambda \bar{x}_W$  and standard deviation  $\lambda s_W$ , so the relation

$$\frac{z - \bar{z}_W}{s_W(z)} = \frac{x - \bar{x}_W}{s_W(x)} \quad (6)$$

holds identically for every  $\lambda > 0$ , leaving the standardised input unchanged and the classification unable to move. Across all eight cells the two arms differ in 0 states, and the reduction in misclassification is 0.00 percentage points in every cell, in sample and out.

Equation (6) is not specific to reliability shrinkage but holds for any affine transformation of the observable, so the entire class of linear proxy corrections applied upstream of a standardising classifier is annihilated, not attenuated. The recoverable band is empty, not small, and the conclusion is available analytically, before any classification has been run.

### 6.2 Noise in the observation equation

The remaining route is to hand the model the noise instead of the correction, decomposing the emission variance in each state into a state variance plus a known observation-noise variance  $v = (1 - \lambda) \text{Var}(\log RV_M)$  and holding  $v$  fixed while the rest is estimated. On synthetic data with known state variance and known added noise the estimator behaves as intended, in that the free-variance

model recovers the total emission variance and the fixed-noise model recovers the state variance by subtracting  $v$ , with the state means of the two agreeing to ten decimal places.

That correctness is what defeats the exercise. With  $\sigma_k^2 = \max(\text{total}_k - v, 0)$  the emission variance is  $\max(\text{total}_k, v)$ , so the fixed-noise model is the free-variance model equipped with a variance floor at  $v$ , identical to it wherever both states clear that floor. The floor is far from idle, since at the headline scaling it binds in 46,120 of 72,619 windows, or 63.5 percent, and moves 18,311 in-sample states. Even so, the maximum reduction in misclassification is 0.00 percentage points in every cell at every scaling, and where the floor binds hardest the classification degrades.

The reason is that a variance floor cannot sharpen a discrimination. Where both states carry the same known uncertainty, knowing how uncertain an observation is tells a two-state model nothing, because the common term enters both emission densities and cancels in the posterior ratio. Only the floor survives that cancellation, and it widens both densities and pulls marginal observations toward the nearer state mean.

### 6.3 Scope of the negative results

Both results are specific to a discrete-state model with state-independent noise, and a model whose latent variable is continuous does not share the cancellation. In a stochastic volatility state-space model with a log-variance signal equation and a known observation-error variance, the noise enters a filtering recursion and not a two-way likelihood ratio. We have not tested such a model and make no claim about its behaviour.

## 7 Practical implications

### 7.1 Sampling frequency for the convexity adjustment

A variance swap strike converts to a volatility swap strike through a convexity adjustment that depends on  $\text{Var}(\log IV)$ . Under lognormal  $V$  the relation is exact and requires no expansion,

$$K_{\text{vol}} = \mathbb{E}[\sqrt{V}] = \sqrt{\mathbb{E}[V]} \exp(-s^2/8), \quad s^2 = \text{Var}(\log V). \quad (7)$$

The second-order expansion in  $\text{Var}(V)/\mathbb{E}[V]^2$  of Brockhaus and Long [2000], commonly used in its place, departs from equation (7) by ten percent at  $\kappa = 0.182$ . The measured  $\kappa$  in this sample runs from 1.26 to 6.84, and at the upper end of that range the expansion returns a negative strike, so we use equation (7) throughout.

Substituting  $\text{Var}(\log RV_M)$  for  $\text{Var}(\log IV)$  overstates the adjustment by between 5.9 and 134.3 percent at five-minute-equivalent sampling. Since the overstatement is a function of the sampling frequency the practitioner chooses, Table 3 states, for each cell, the frequency at which the overstatement falls below five percent.

One-minute sampling suffices at the daily horizon on both instruments and under both session geometries, but nowhere else. At the thirty-minute horizon on either instrument, and at the hourly horizon on ES, no frequency available in the grid brings the bias under five percent. A thirty-minute window does not contain enough returns for the proxy's own dispersion to fall below the quantity being measured. A practitioner pricing a volatility swap on a thirty-minute horizon cannot remedy the bias by sampling faster.

Table 3: Sampling required for the convexity bias to fall below five percent.

| Cell           | Bias at five-minute sampling | Sub-bar length required |
|----------------|------------------------------|-------------------------|
| ES/GLOBEX 1day | 19.3%                        | 1 minute                |
| NQ/GLOBEX 1day | 5.9%                         | 1 minute                |
| ES/RTH 1day    | 17.6%                        | 1 minute                |
| NQ/RTH 1day    | 6.9%                         | 1 minute                |
| NQ/RTH 1h      | 21.0%                        | 1 minute                |
| ES/RTH 1h      | 62.6%                        | unreachable             |
| ES/RTH 30min   | 134.3%                       | unreachable             |
| NQ/RTH 30min   | 42.1%                        | unreachable             |

No options data is held, and we report a bias in the adjustment term without any claim about executable profit.

## 7.2 Misclassification at a median threshold

Where a decision turns on a threshold placed inside the distribution rather than in its tail, the misclassification rate follows from reliability alone. With thresholds at the respective medians and lognormal noise, the correlation between  $\log RV_M$  and  $\log IV$  is  $\sqrt{\lambda}$ , and the bivariate normal orthant probability gives

$$\Pr[\text{disagree}] = \frac{\arccos\sqrt{\lambda}}{\pi}. \quad (8)$$

Equation (8) is exact under those assumptions and is plotted in Figure 2. A practitioner who has measured  $\lambda$  for their own series can read the implied rate off the figure directly.

The measured cells sit between 7.9 and 13.6 percent, while the empirical rate measured against a finest-grid noise-robust proxy is lower, falling between 50 and 77 percent below the analytic value. The two series are computed from the same window and their errors are positively correlated, so the empirical figure should be read as a lower bound. Moving the classification boundary from the median to the two tercile boundaries raises the empirical rate to between 6.0 and 11.9 percent. The mechanism there is the count of boundaries, not the density at them, since two boundaries admit more disagreements than one. For a unimodal distribution the tercile boundaries sit further from the mode than the median does.

## 7.3 Volatility-target sizing

We recommend against correcting volatility-target sizing for measured reliability. On the holdout, replacing the textbook shrinkage weight with the measured one changes tracking error by at most 1.008 percent in relative terms, in the same direction in every cell and at every point of a cost sweep spanning an eightfold range of transaction cost. Reliability at five-minute-equivalent sampling runs from 0.40 to 0.94 against a textbook 0.67 to 0.99, so the gap being corrected is real and large, but it does not move a position far enough to matter at a ten percent volatility target with daily rebalancing. The same holds for inverse-volatility risk parity at monthly rebalancing, where averaging the estimate over twenty-one sessions divides the bias term by twenty-one.

## 8 Limitations

**The composition of  $AM^b$ :** Equation (5) attributes the whole excess over the fitted intercept to proxy error, whereas direct measurement of that error against a finest-grid realized kernel gives between 9.1 and 41.8 percent of the same quantity. The fitted term is an upper bound on the noise and the kernel difference a lower bound, and the two differ by a factor of 2.4 to 11, and every  $\lambda$  we report should be read as a lower bound. Resolving the composition would require measuring proxy error against an instrument that does not share the window, which we have not done.

**Two correlated instruments:** The futures evidence is ES and NQ, two equity index contracts on the same exchange whose realized variances are strongly related, and the independent weight in the cross-sectional evidence is carried by SPY alone, at two venues under traded-tick sampling. Where a result is said to hold in 18 cells, 16 of those cells are two instruments observed in eight configurations.

**Cluster saturation:** Every trend  $p$ -value in this paper comes from a wild cluster bootstrap with Rademacher weights and the null imposed [Cameron et al., 2008], clustering on eight distinct cells. At  $G = 8$  there are  $2^7 = 128$  distinct values of the statistic, and the smallest attainable two-sided  $p$ -value is 0.0078. The reported values of 0.0076 and 0.0036 sit at or below that floor, so the test is saturated and none of these values constitutes conventional evidence at any nominal level. The design does support the point estimate and its sign, which is  $-0.0469$  per year uncontrolled and  $-0.0105$  per year once fit conditioning enters, with a variance inflation factor of 1.27 on the year term. Four fifths of the uncontrolled trend is attributable to fits becoming less well identified in high-volatility years.

**Holdout openings:** The holdout was opened seven times, for sizing and the candidate set out of sample, re-evaluation of three candidates after a predictor-construction defect; risk limits and combination weights; risk parity; regime misclassification; regime classification under an observable-side correction; and measurement error in the emission. No parameter, threshold, rule or specification was changed after any holdout number had been seen, and each opening was pre-registered before the read. The programme is not single-use on the holdout, and we do not describe it as such.

**The thirty-minute inversion:** One cell runs 48.5 percent misclassification in sample against 26.5 percent out of sample, which inverts the usual direction, since performance at chance in sample is followed by performance better than chance on the holdout. Label instability under close state means is present, and is located where it would be expected. In sample, 17.4 percent of windows carry a state-mean gap below 0.25, against 6.9 percent in the holdout. Restricting attention to well-separated windows nonetheless moves the two rates by only 0.67 and  $-0.04$  percentage points, about three percent of a 21.95 point inversion, and mean state separation is 1.709 in sample against 1.668 out. Neither candidate accounts for the inversion, and we leave it unexplained.

**Partial mechanism:** Calibrated within-window volatility dispersion reproduces the observed exponent on GLOBEX geometry at every Hurst index tested and on RTH geometry at some. Three candidate mechanisms have been tested against the RTH residual and rejected. Independent heavy-tailed innovations account for 8.6 percent of the gap and lie inside seed noise; a fixed-position

dominant return moves the exponent nowhere, at any position or amplitude; and the roughness of the within-window path has a total range across Hurst indices smaller than its own between-seed dispersion. A localized feature would need to carry between 16.1 and 32.3 percent of window realized variance to supply the observed floor, against measured shares between 0.41 and 4.04 percent. That is a shortfall of 4.0 to 71.5 times in share and, since the floor goes as the square, of 16 to 5,100 times in the quantity supplied. The class of localized mechanisms can be retired, and the source of the RTH residual is not established here.

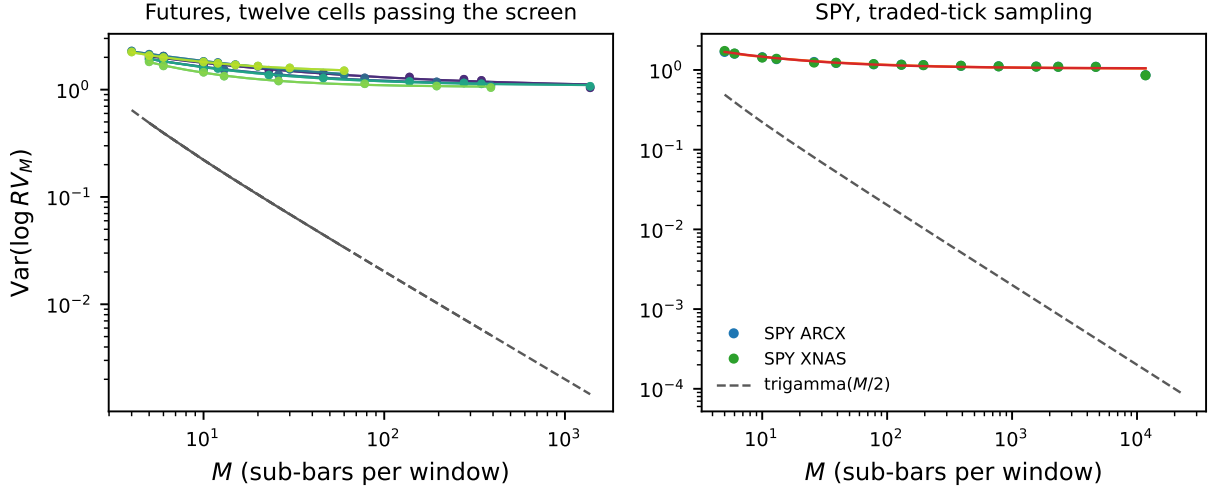


Figure 1:  $\text{Var}(\log RV_M)$  against  $M$  on log axes. Left: the twelve futures cells passing the tightened screen, points read from `s08-final/results/phase4_lambda.csv` and fitted curves from `s10-exponent-audit/results/phase1_bootstrap.csv`. Right: both SPY venues under traded-tick sampling, points from `s07-completion-and-spy/results/phase6_spy_grid_{ARCX,XNAS}.csv`. Dashed lines are  $\text{trigamma}(M/2)$  on each grid. Twelve futures cells and two SPY venues pass the screen, fourteen in total; the twelve futures cells are six distinct cells under two boundary treatments that give identical fits.

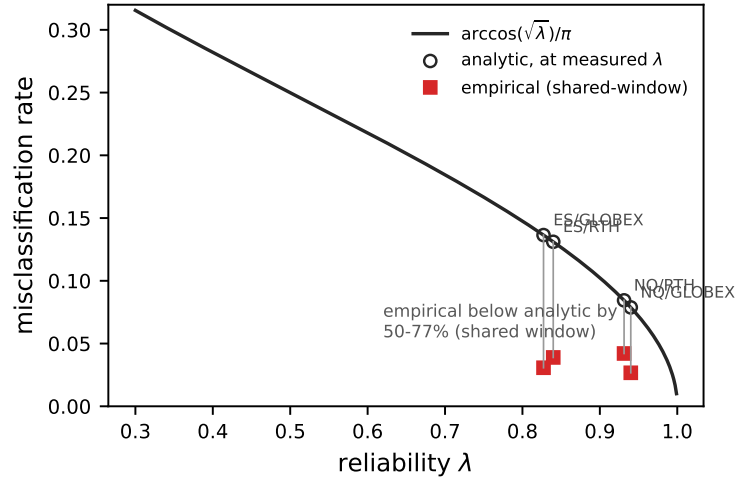


Figure 2: Misclassification against reliability. The curve is  $\arccos(\sqrt{\lambda})/\pi$ . Open circles mark the analytic rate at each cell's measured  $\lambda$ ; squares mark the empirical rate against a finest-grid noise-robust proxy, which is a lower bound because the two series share a window. Read from `s14-applications/results/phase1_k8_rates.csv` and `s09-application/results/phase3_sizing_params.csv`.

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